SLIC (SIMPLE LINE INTERFACE CALCULATION)*

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Abstract

SLIC is an alternating-direction method for the geometric approximation of fluid interfaces. It may be used in one, two, or three space dimensions, and it is characterized by the following features: (1) Fluid surfaces are represented locally for each mixed- fluid zone, and these surfaces are defined as a composition of one space dimensional components, one for each coordinate direction. (2) These one-dimensional components are composed entirely of straight lines, either perpendicular to or parallel to that coordinate direction. (3) The one-dimensional surface approximations for a mixed fluid cell are completely determined by testing whether or not the various fluids in the mixed cell are present or absent in the zone just to the left and to the right in the coordinate direction under consideration. (4) Because of the completely one-dimensional nature of the SLIC interface description, it is relatively easy to advance the fluid surfaces correctly in time. With the SLIC fluid-surface definitions, it should be possible to incorporate any one space dimensional method for advancing contact discontinuities. This makes SLIC very practical for the numerical solution of fluid dynamical problems.

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Introduction

This paper deals with the problem of treating fluid interfaces in the context of multifluid Eulerian hydrodynamic calculations. The problem presents two fundamental difficulties, that of defining geometrical approximations to the fluid interfaces and that of formulating essentially Lagrangian equations of motion to advance these fluid interfaces correctly in time.* Our discussion will focus on the former difficulty, although the simplicity of the SLIC surface approximations lends itself to a solution (see [2]) of the latter difficulty as well.

Of the many approaches to calculating multifluid flow in a two-dimensional Eulerian context, three methods are particularly noteworthy: (1) the particle-in-cell (PIC) method [4], which uses mass particles to tag and keep track of the various fluids; (2) the coupled-Eulerian-Lagrangian (CEL) method [1], which explicitly uses Lagrangian polygonal lines to define fluid interfaces; and (3) the highly successful, though less well publicized, KRAKEN method [5], in which fluid interfaces are defined locally for each mixed-fluid zone.

In the method to be described here, we have followed the KRAKEN approach in which a local surface approximation for each mixed-fluid zone is defined. However, we have sought the simplest possible interface description, and this has led us to an alternating-direction method, referred to as SLIC (simple line interface calculation). Roughly, the SLIC rule for approximating surfaces is that lines perpendicular to the coordinate axis are to be preferred to parallel lines and that for three fluid zones in which (in two dimensions) surfaces meet in a "Y like" intersection, then that intersection is to be approximated by a "T". Also a few symmetry considerations are invoked to ensure that fluids get equal treatment when the same information is known about each.

Surprisingly enough, the limited information obtained merely by looking to the left and to the right in each coordinate direction is sufficient, when coupled with the simplicity of the straight-line surface approximations, to construct a reasonable and workable representation of all possible fluid surface configurations of any number of different fluids occupying a mixed-fluid cell. The alternating-direction feature of SLIC makes it easily generalizable to more than

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*See [1], [2], [3] for examples and a more detailed exposition of the problems associated with the correct numerical approximation of contact discontinuities.